Smart Watering Device

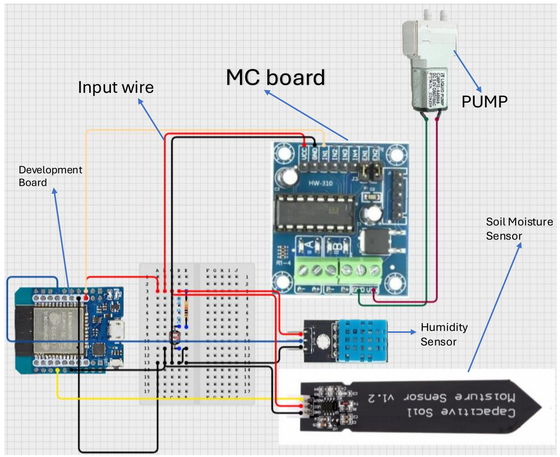
Group 03: Alvaro Quequejana, Declan Best, and Joshua BeckerCaceras

ENGR 111: Intro to Engineering Design – Fall 2025

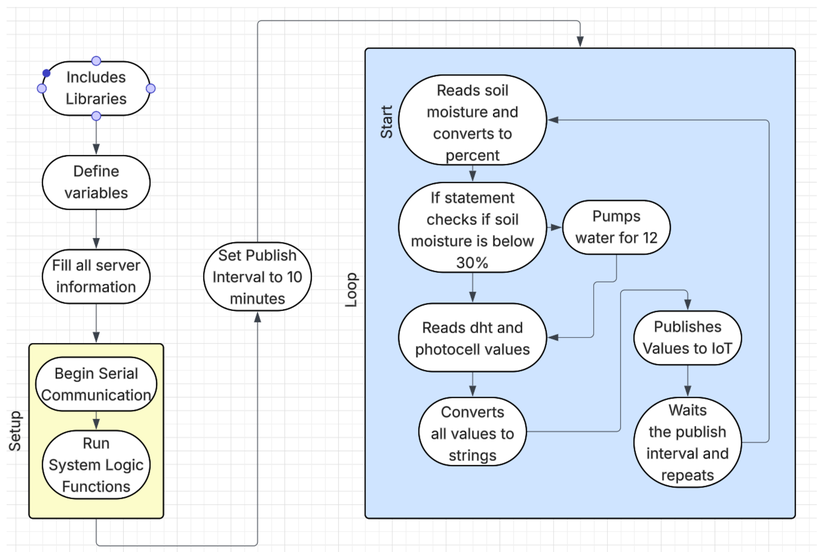
PROBLEM STATEMENT

Manual plant watering can be inconvenient and inconsistent, resulting in unhealthy plants. Our solution is a smart watering system that automatically waters the plant based on soil moisture levels. We focused on keeping electrical components safe from water and its integration to a water source.

Circuit Diagram



Software Flow Chart

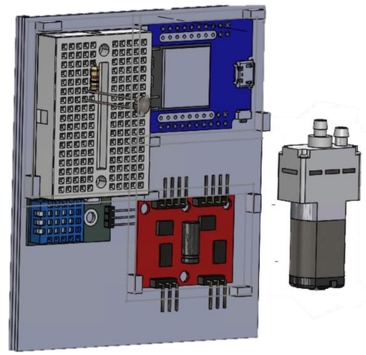


DESIGN OVERVIEW

Electrical:

- The wires fell out very easily so we made the area smaller.
- In order to make room for the photocells output we had to add an extra wire so it could be separate and then attach it to the ground line.
- There also were not enough voltage pins so we used the breadboard to spread it across all the devices.

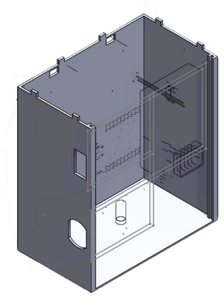
Software: See Software Flowchart on bottom left.



Motherboard (The Sliding sheet holding the circuits & parts)

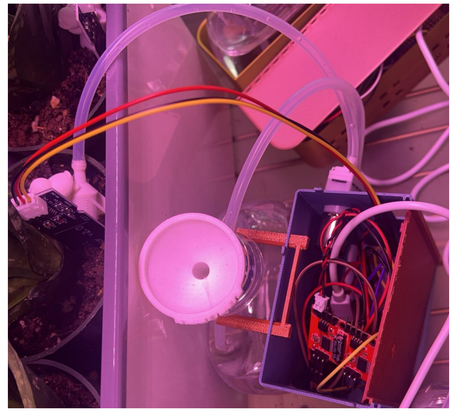


Base Dimensions: 3.5" x 2 " x 4"



The Holder (Where Motherboard slides in to contain and protect)

SYSTEM TESTING



Our prototype in action!

Field Measurements:

